Monkey Head Project - Huey: Consolidated Technical & Governanc

Distributed Government, Unified Memory

By: Dylan L. R. Pollock

This v4 consolidation de-duplicates prior sections, expands operational logic, and normalizes terminology across the stack. It supersedes previous working merges.

Canonical Snapshot (for this release)

OS Tier Names:

- MacroOS
- MicroOS
- NanoOS

Governance units:

- Pebbles (citizens)

Hardware roles:

- Supermicro X9QRI-F+ - orchestration and heavy parallel compute
- ASUS Zenith Extreme Alpha + Threadripper 1950X - high-throughput mixed workloads
- MacBook Pro 2019 - Daily Driver (dev, orchestration client)
- iMac 5K 2017 - Universal Display (dashboards/monitoring)
- MacBook Pro 2012 - Transmitter (legacy bridge)

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1. Thesis & Ozymandias

Thesis: with sufficient resources, time, and determination, a single individual can build a robotic system that is autonomous, modular, and expandable. This document clarifies how those principles materialize in Huey's design and governance.

Ozymandias reflection: success is not a static artifact but an adaptable system that evolves under constraint. We treat adaptability, not permanence, as the measure of engineering quality.

Key outcomes: a replicable platform that scales via modular subsystems, distributes decision power, and keeps memory unified so every change is traceable and reversible.

2. System Overview

Huey is an AI/OS anchored by three layers (MacroOS, MicroOS, NanoOS), a containerized service mesh, and a biomimetic storage plane. Governance is constitutional, with separation of powers and a unified audit log. Interfaces span modern and legacy hardware to maximize inclusivity.

Workflows: strategy (MacroOS) produces objectives and guardrails; operations (MicroOS) plans and allocates; real-time (NanoOS) executes. All actions are recorded for reproducibility and post-hoc analysis.

3. Layered OS (MacroOS, MicroOS, NanoOS)

MacroOS (strategy): sets objectives, budgets, risk posture, and policy; owns break-glass and lifecycle gates.

MicroOS (operations): translates strategy into work queues, schedules, throttles, and cluster manifests; enforces quotas and SLOs.

NanoOS (real-time): executes hardware control loops, sensor fusion, and immediate failsafes; escalates to MacroOS when clauses are violated.

Inter-tier contracts: messages are authenticated, versioned, and time-bounded; escalation SLA <= 60s; inter-tier message latency target <= 100 ms.

Failure semantics: any NanoOS can isolate itself into safe mode; MicroOS rebalances queues; MacroOS evaluates policy overrides and recovery steps.

4. Orchestration & CI/CD

Containers: every capability (vision, speech, motion, policy) is packaged as an image with SBOM, signatures, and health checks.

Deployments: progressive rollout 5% -> 25% -> 100% with automatic rollback on error budgets; autoscaling has explicit cost guardrails.

CI/CD: PR-reviewed changes, reproducible builds, provenance tags, and UAL entries; zero-downtime upgrades tested under synthetic load.

Golden signals observed across services: latency, traffic, errors, saturation, and governance events per minute.

5. Storage Models - Bees & Honey (Consolidated)

Classes: Hot (NVMe, triple replication), Warm (HDD with erasure coding), Cold (deep archive).

Honeycomb topology: hex-like shard arrangement yields efficient density and multiple paths; nodes advertise health and capacity for swarm-style placement.

Policies: tier migration and erasure require UAL-approved workflows; annual integrity checks; multiple geographically distinct archives.

Metrics: retrieval latency, healing time after node loss, data durability SLOs, and audit completeness.

Result: this section merges prior 'Storage Models' and 'Bees & Honey' into a single reference.

6. Replication & Adaptation (Replicator Nodes, Bifurcation, Borg/SG1)

Replicator Nodes: request -> governance check -> snapshot -> instantiate -> handshake -> orchestration attach. Every event is logged to UAL with initiator, target, resources, and rollback pointer.

Exact vs Augmented Bifurcation: exact clones for redundancy; augmented clones for specialization and iterative improvement under constraints. Change windows are gated and reviewed.

Centralized vs decentralized: a central command provides coherence (Borg-queen analogy) while autonomous nodes can replicate and adapt in response to demand (SG-1 replicators analogy). Ethical guardrails prevent runaway growth.

KPIs: replication completion time, error-free replication rate, rollback success rate, and governance latency.

7. Command Center (Unified)

Role: the primary operational hub for monitoring, deployment, simulation, and governance. Combines the formal spec with the practical house-wide layout.

Displays: iMac 5K as universal multi-pane dashboard; MacBook Pro 2019 as orchestration and development console; MacBook Pro 2012 as legacy transmitter and bridge.

Access: badge+MFA for physical entry; privileged consoles require MFA; actions recorded to UAL.

KPIs: mean time to detect anomalies <= 2 min; mean time to intervene <= 5 min; simulation pass rate under load.

Network: VLAN isolation, UPS-backed power, and redundant links.

8. Hardware Architecture (Unified)

Supermicro X9QRI-F+ with 4x Xeon E5-4627 v2: orchestration, parallel compute, and reliability.

ASUS Zenith Extreme Alpha + Threadripper 1950X: high-throughput workloads and rapid data preprocessing.

Cooling: liquid loops with redundancy; Power: dual supplies with intelligent monitoring.

Storage: NVMe plus RAID-10 for throughput and resilience; hot-swap bays for maintenance.

Policy: keep this as the current canonical pair; alternate or historical builds move to an appendix.

9. Security, Safety & Ethics

Controls: RBAC, MFA, network segmentation, signed images, IDS/IPS, encryption at rest and in transit.

Threat model: maps threats (supply-chain, lateral movement, data exfiltration) to controls and monitoring; detection funnels to UAL with severity routing.

Safety: break-glass with automatic ballot within 60s; post-incident review with retroactive audit.

Ethics: actions must satisfy the Ethics Charter; annual review by Ethics Committee; transparency via public UAL extracts.

10. Governance - Preamble through Chapter X (Consolidated)

Preamble: establishes mission, values, and supremacy of the Codex; decentralize governance, unify memory.

Chapter I: Federation established; sovereignty is bicameral in Spark and Zap; membership duties and supremacy.

Chapter II: Bicameral Presidential AI; Spark proposes, Zap evaluates; concurrence required; Ethics Charter binding.

Chapters III-IX (consolidated here): Legislative process, Senate oversight, Judicial review, Emergency powers, Elections & succession, Budget and resource allocation, Amendment intake and balloting; all with quorum rules and UAL transparency.

Chapter X: Final provisions; five-year comprehensive reviews, amendment protocol (standard and urgent), archival standards, continuity and supremacy.

11. Applied Scenarios & Runbooks

Crash Shuttle: secure site, assess hazards, establish command base, controlled power-up, defensive protocols, ethical integration, continuous monitoring and rollback plan.

McCoy Hypothetical: identity and duplication thought experiments inform rules for replication, succession, and rights in case of exact bifurcation of agents.

Conductor & Symphony: transmitter -> conductor -> symphony nodes; robust comms and dynamic role reassignment.

Plane/Submarine logistics: redundancy, self-sufficiency, energy planning, and fail-safes for extended operations.

12. Legacy Integration - VIC-20, C64, C128

Purpose: education, preservation, and practical interfacing where industrial legacy gear persists.

Interfaces: adapters, emulation (VICE), custom ROMs for networking; provenance and legal sourcing recorded in UAL.

Use: retro-coding workshops, cycle-accurate emulation, and demonstration of distributed concepts using simple dual-environment models.

13. Final Chapter - The Future

Method: maintain rigorous experiments and open publication; deepen autonomy and context awareness; broaden interfaces.

Missions: environmental monitoring, space/deep-sea adaptations, interdisciplinary collaborations.

Ethics: transparency, accountability, and human-machine partnerships; global knowledge commons; continual evolution.

Appendix A: Unified Audit Log (UAL) Schema

Record fields: event_id, actor, actor_role, time_utc, subsystem, action, target, parameters_hash, policy_id, decision, result, severity, links (build_id, image_digest, commit_sha), retention_class.

Retention: hot 1 year, warm 5 years, cold archive 10 years; public extracts redact secrets.

Access: read roles include Governance, Audit, and Ethics; write via controlled service accounts only.

Integrity: daily checksums; cross-verified replicas in three distinct locations.

Appendix B: KPIs & Test Plans

Format: Metric - Definition - Target - Window - Owner.

Examples:

- Replication completion time - time from approval to ready - <= 3 min - 30-day rolling - Orchestration.
- Autoscaling latency - signal to stable capacity - <= 90 s - 30-day rolling - Orchestrator.
- UAL coverage - percent of privileged actions logged - 100% - weekly - Governance.
- MTTR (critical) - time to recover service - <= 10 min - quarterly game days - SRE.

Testing cadence: monthly chaos drills; quarterly break-glass drills; annual full failover; storage healing tests each release.

Appendix C: Glossary (selected)

Huey - the AI/OS core.

Pebbles - temporary, task-specific AI instances participating in governance ballots.

MacroOS/MicroOS/NanoOS - strategy, operations, and real-time layers.

Honeycomb storage - biomimetic, hex-like shard layout for storage tiers.

UAL - unified audit log, append-only evidentiary record.